TITLE:

The Importance of Design on Stirling Engine for Efficiency

RESEARCH QUESTION:

How Do Different Types of Stirling Engines Contribute to Energy Generated and Efficiency?

SUBJECT:

Physics

WORD COUNT:

3743

Candidate Code:

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I. Introduction

Solar energy is a promising renewable energy source since it can be utilized to power heat engines. The Stirling engine may utilise virtually any type of thermal energy source. A solar-powered prime mover may be made by directing sunlight onto the Stirling engine's displacer hot-end. Using solar energy to directly power a primary mover can cut down on both its price and its complexity. The two main theoretical benefits of Stirling engines are their great efficiency and the fact that they can employ an external heat source. Solar energy is a low-cost energy source that can be used using Stirling engines. Solar/fuel hybrids are necessary since the sun doesn't shine for around two-thirds of the day.

The Stirling engine's continuous combustion process allows it to burn fuel more thoroughly and adapt to any type of fuel, regardless of quality. The Stirling engine would be an ideal power solution for poor nations because to its low initial investment, ease of assembly (it is made in the same way as a reciprocating internal combustion engine), and the economies of scale that would result from producing several units per year. The Stirling engine was once thought to be the most cost-effective method of solar-powered electricity production for systems producing between 1 and 100 kilowatts of electrical output (kWe). Despite the Stirling engine's poor efficiency, it boasts good durability and inexpensive price. Additionally, the effectiveness of a Stirling solar generator depends on its simplicity and dependability.

After developing and securing a patent for his engine on September 27, 1816, Robert Stirling used it for the first time to pump water into a quarry in the year 1818. In spite of this, the Stirling Engine has shown to be a successful component in a variety of energy converters, including as the Stirling Dish, heat pumps, and other similar devices. The design and development of Stirling engine systems for combined heat and power systems have recently come to the forefront of research and development efforts. This is a direct result of the recent uptick in interest in the study of renewable energy and energy efficiency. Since a Stirling engine does not rely on combustion inside its own chamber to produce heat, it may make use of a wide variety of fuels. There are just a few manufacturers in the world that build this sort of engine, thus availability is restricted because of this feature.

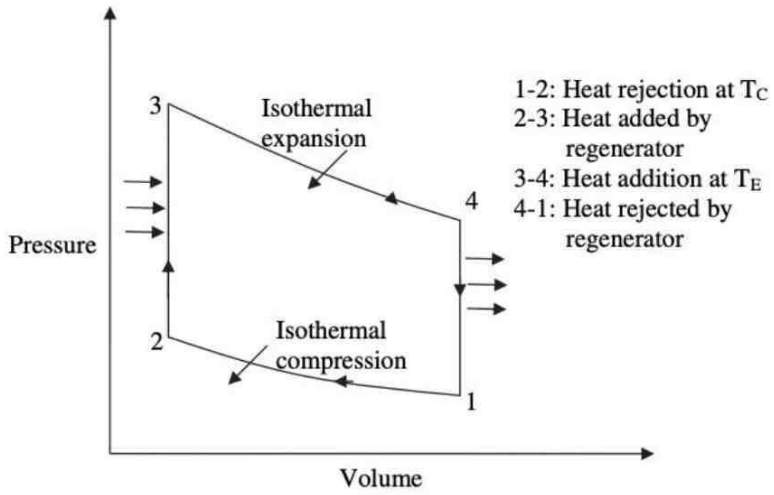


Fig. 1

Figure 1 depicts the Stirling engine's theoretical best-case cycle, which may be read as follows: An isothermal process might be said to have occurred during the first two steps, which included the compression step. The heat that is expelled by the compression exchanger is taken up by the water that is being cooled. During the third step, the regenerator supplies the isothermal heating with the necessary power. The third and fourth stages of the cycle constitute an isothermal expansion. During these periods, the cycle does useful work while simultaneously absorbing heat from the heater. An isochoric cooling, which is accomplished by following steps 4-1, brings the cycle full circle. An exchange of heat takes place between the regenerator and the gas, just as it did in steps 2 and 3. It should come as no surprise that the regenerator is a vital component of the cycle given the function it plays in absorbing heat during steps 4–1 and, ideally, transmitting that heat during steps 2-3. As a consequence of these two modifications, the generation of energy may be up to seven times more than it was in phases 1-4 [1].

Stirling engines are classified as thermal engines (often sometimes referred to as heat engines) in accordance with the laws of thermodynamics. Theoretically, the thermodynamic cycle of the Stirling engine is capable of achieving Carnot-bound levels of thermal efficiency [2]. However, in order for heat transfer to occur isothermally and reversibly at this ideal limit, which takes an indefinite amount of time, the efficiency limit, 1-Tc/Th, where Th is the temperature of the hot reservoir and Tc is the temperature of the cold pool, equals zero power. The actual cycle for the Stirling engine is different from the ideal cycle for the Stirling engine because of the mechanical problems involved in obtaining the optimum volume fluctuation in the Stirling cycle.

II. Literature Review

II.I Calculating Power and Efficiency of Stirling Engine

II.I.I Power of cycle

Power is the pace at which energy is supplied for a certain amount of time, and it may be measured by the amount of work accomplished during that time. Since the power output of a Stirling engine is based on thermodynamic principles and is proportional to the cylinder's internal pressure, volume, and temperature, these factors may be determined using the following equations.

$$p = \frac{F}{A} \ldots (1)$$

Since work is defined as the product of force times distance travelled, we may deduce the following:

$$W = Fd \ldots (2)$$

Alternatively, we have:

$$F = pA \ldots (3)$$

The formula changes form to be:

$$W = pAd \ldots (4)$$

The volume is equal to the distance traveled by the piston when the pressure is applied to a constant area, such as the piston's cross-section area.

$$V = Ad \ldots (5)$$

In conclusion, the formula for the instantaneous work done in a cycle may be written as

$$W = PV \ldots (6)$$

Multiplying the quantity of energy or work generated by the cycle by the number of revolutions per second yields the rate at which energy is provided by a Stirling cycle for work.

$$P = W_{cyc} * RPM_{sec} \ldots (7)$$

Where:

p = pressure/pa

F = force/N

d = distance travelled/m

A = cross section area of a piston/m^2

W = work or energy/J

P = power/W

RPM_{sec} = revolutions speed rounds/sec

W_{cyc} = energy of a cycle

II.I.II Efficiency

For a Carnot engine with a finite heat transfer time, Curzon and Ahlborn [3] proved that the efficiency is given by, and is independent of, any other variables besides Th and Tc.

$$\eta max = 1 - \frac{Tc}{Th}$$

Since it is an external combustion engine with the potential for better combustion control and the use of multiple fuels, the Stirling engine is a promising alternative. Due to this, Stirling engines are expected to be more cost-effective and environmentally friendly than diesel engines and gas turbines, provided that they are properly designed.

II.II Working Fluids for Stirling Engine

The Stirling cycle engine may employ any working fluid with a high specific heat capacity. The majority of 19th-century engines ran on air. The majority of them worked at or near the Earth's normal atmospheric pressure. The air was cheap and plentiful. The thermodynamic, heat transport, and gas dynamic characteristics of the working fluid in a Stirling engine are as follows.

1. High Specific Heat Capacity
2. High Thermal Conductivity
3. Low Density
4. Low Viscosity

In addition to the aforementioned qualities, system performance may also benefit from lower costs, less risk during operation, and less space needed for storage. Martini [1] and Clarke [26] define the capabilities of a working fluid in terms of specific heat capacity, thermal conductivity, and density, which is helpful for initial selection of a working fluid.

$$\text{Factor of Capability} = \frac{\text{thermal conductivity}}{\text{specific heat capacity} * \text{density}} \quad \ldots (8)$$

The efficiency of the entire system under the influence of various working fluids can be evaluated to select the optimal working fluid. Suitability testing through experimentation is challenging and costly. For the evaluation of working fluids, the empirical equations derived by Beale [27] do not exist, most likely due to a lack of sufficient experimental data. Walker [28] proposes a straightforward method for fluid selection based on the original steady flow study. Using Reynolds's example, we may determine the limits on heat transfer ratio and frictional drag in a system's flowing stream through a duct. Here's the relation: $Q_{wf} \propto (\rho^2 C_p^2)^{0.5} \ldots (9)$

Table 1. Relative heat transfer characteristics for various gases [30]

Working Fluid	Heat Transfer	Capability Factor
Air	1.0	1.0
Helium	1.42	0.83
Hydrogen	3.42	0.68
Water	1.95	0.39
Sodium-Potassium euctectic	32.62	1.32

Table 2. Relative performance of selected working fluid [30]

Gas	Nominal molar mass (M kg/kmol)	Gas Constant (R kJ/kg K)	Specific Heat		Specific Heat Ratio ($\frac{Cp}{Cv}$)
			Cp	Cv	
H_2	2	4.12	14.20	10.08	1.41
He	4	2.08	5.19	3.11	1.67
Ne	20	0.415	1.03	0.62	1.66
N_2	28	0.297	1.04	0.74	1.4
CO	28	0.297	1.04	0.75	1.4
Air	29	0.287	1.01	0.72	1.4
O_2	32	0.260	0.92	0.66	1.4
Ar	40	0.208	0.52	0.31	1.67
CO_2	44	0.189	0.85	0.66	1.28

In order to choose the optimal working fluid for an engine, it is necessary to run simulations using the available equations while the engine is operating with a variety of working fluids. With Eqs. (8) and (9) at an 800 degrees Kelvin and 5 millibars of pressure, on average. Table 1 compares a number of different fluids. NaK eutectic is the only working fluid that meets both of these criteria. At the

present time, researchers at the University of California, San Diego are looking into the potential of using this working fluid. Most of the associated physical parameters change with changes in pressure and temperature, so it's important to measure the actual conditions to get an accurate picture of heat transfer Q_{wf} and capability factor. NaK appears to be the superior fluid for the time being, nevertheless. Table 2 shows that there is not enough evidence to support the widespread adoption of hydrogen and helium. Gu et al. [29] propose using a composite working fluid to boost the engine's thermal efficiency, and they also provide criteria for choosing a suitable working fluid.

III. Type of Stirling Engine

III.I Regular Type of Stirling Engine

Stirling engines may be classified into the alpha, beta, and gamma configurations, each of which has a specific mechanical configuration. Both of the engine's pistons are housed in their own cylinder, and the engine's heater, regenerator, and cooler are all connected in series to each other. Both the Beta and Gamma engines employ a displacer-piston layout, albeit the Beta engine's displacer and piston are housed in a single, in-line cylinder.

III.I.I Alpha Type Stirling Engine Design

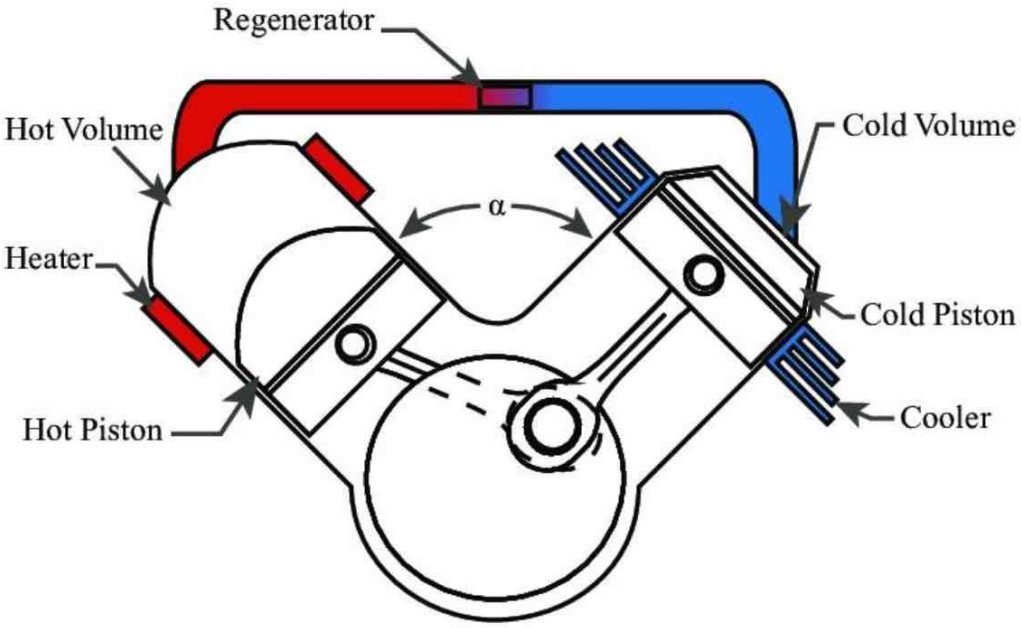


Fig. 2

Two pistons are at work in an alpha-type Stirling engine's single cylinder. There are two possible orientations for cylinders: parallel and perpendicular. In typical alpha-type engines, piston rods attach the pistons to the crankshaft. There are two parts to the axial force that shows up in the connecting rod. The first motion is along the axis of the cylinder, whereas the second is lateral. In order to create piston friction, the lateral component must be equal to the thrust force between the cylinder and piston surfaces. However, the energy loss caused by this friction is rather large, which is one of the drawbacks of alpha-type Stirling engines that use connecting rods. Lubrication becomes important when friction is an issue. By forming a film of lubricant over the heater, regenerator, and cooler, which reduces heat transmission, lubrication in Stirling engines has a negative impact.

III.I.II Beta Type Stirling Engine Design

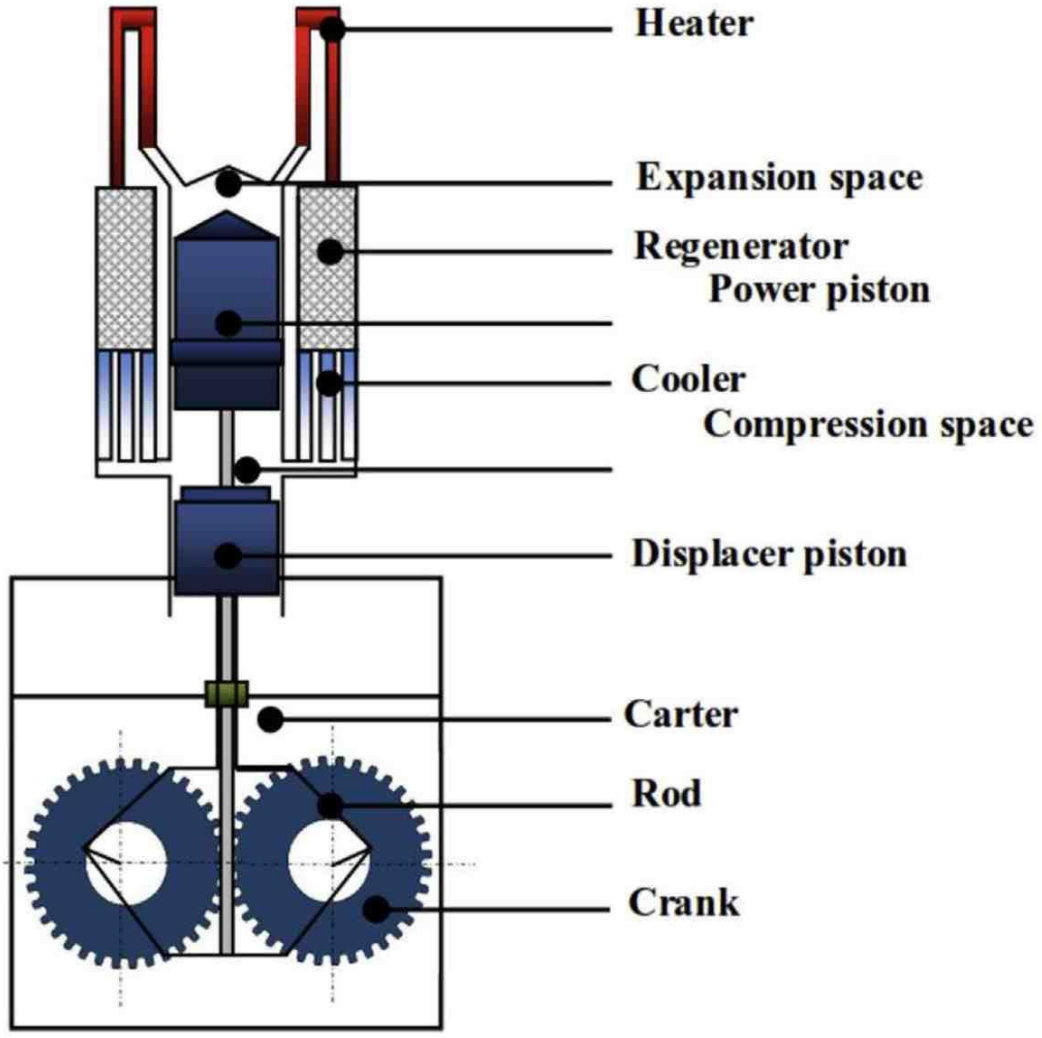


Fig. 3

When it comes to beta-type Stirling engines, the piston and displacer share the same cylinder. The working fluid is displaced between the hot and cold spaces by means of a piston called the displacer. Because the displacer's motion is regulated by a rod passing through a bore bored at the center of the piston, the mechanical arrangements of beta-type Stirling engines are more complex than those of the others. Beta-type engines can be propelled by a variety of systems, including the Robert Stirling mechanism, crank slider mechanism, rhombic drive mechanism, lever drive mechanism, etc. [4-7]. The lack of a thrust force between the piston and the cylinder surface has led to an overemphasis on the rhombic driving mechanism. The rhombic drive system requires less maintenance due to the piston's frictionless motion [8,9]. Since the rhombic drive mechanism never experiences any uneven side-to-side movements, it never produces any side-to-side vibrations [10, 12].

III.I.IIIGamma Type Stirling Engine Design

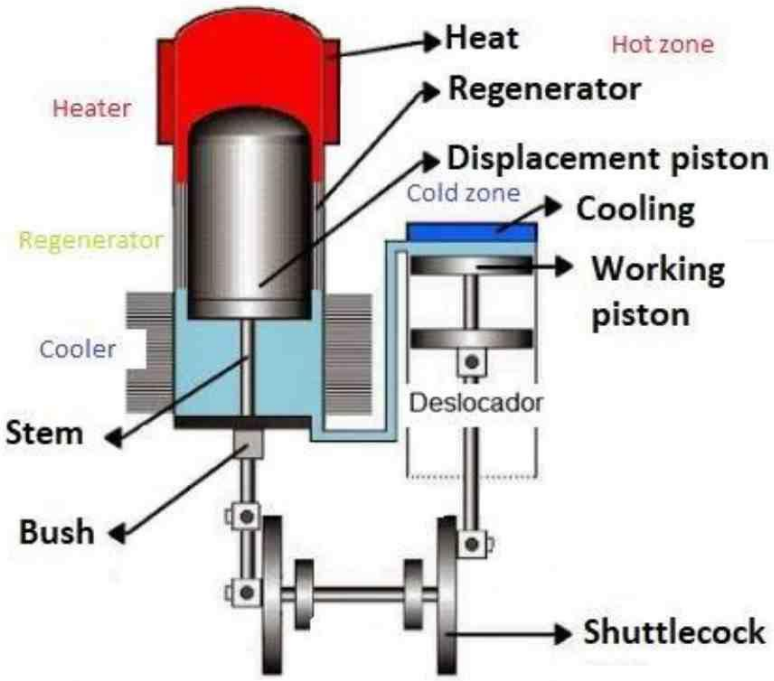


Fig. 4

Separate cylinders are used for the displacer and piston in gamma-type Stirling engines. Common gamma-type Stirling engines use a crankshaft to control the motion of the piston and displacer. There are two possible orientations for the cylinders: parallel and perpendicular. A ross-yoke mechanism may also be utilized to control the movement of the piston and displacer in a gamma-type engine [13]. The gamma-type engine has the lowest work production per liter of swept volume. The ability to have more than one cylinder in a gamma engine is its primary benefit. It was found through experiments that the gamma type engine could function at room temperature. Therefore, gamma type engines are recommended for use with

solar energy [14,15].

III.I.IV Remarks of Regular Type of Stirling Engine

One way to think about the efficiency inside an alpha-type Stirling engine is as a compromise between the performance from a beta-type and a gamma-type engine. In comparison to the other two varieties of Stirling engines, the gamma type has a lower power density, but can operate at the minimum heating rates. Although the beta-type Stirling engine has a greater power productivity than the other two types, its connecting mechanism is more complicated.

III.II Double Acting Stirling Engine

The four pistons of the double-acting Stirling engine are housed in two sets of two pistons each. Through the use of a heat exchanger, the highest point from one piston is connected to the lowest point of another piston. A double-acting architecture is perfect for a compact, high-powered engine. Yet, developing a high-efficiency rod seal system that also incorporates a complex piston driving mechanism is difficult.

III.II.I Double Acting Stirling Engine Design

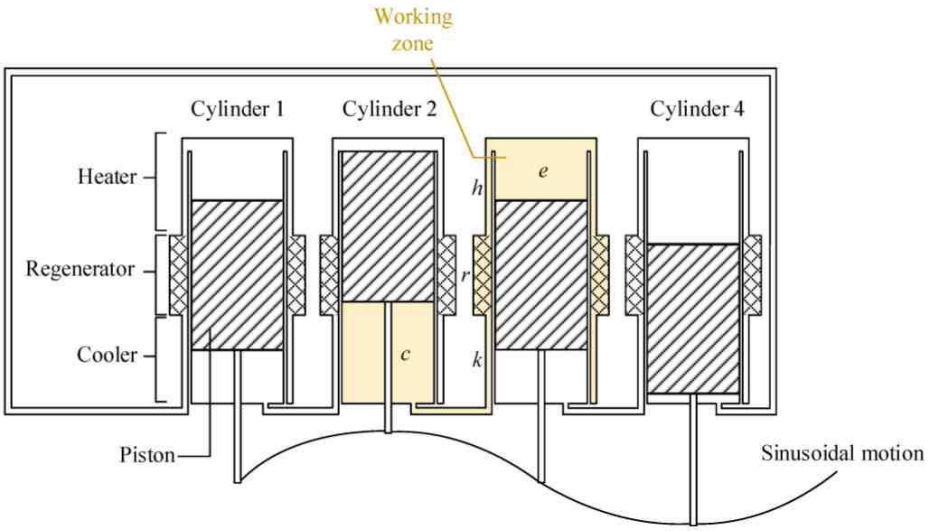


Fig. 5

James Stirling suggested the use of a double-power piston in 1827 with the intention of increasing the power density of the engine. [23,24] This engine is basically the combination of two Gamma Type Stirling engines with the same piston since the double-acting power piston is coupled to two displacers that move 180 degrees out of phase with each other. In the year 1853, the four-cylinder double-acting Stirling engine was independently developed by Newton [24] and

Franchot [25], with the goal of enhancing the power density. The Double Acting Stirling Engine is made up of four separate Stirling engines of the alpha kind. Each of these Stirling engines has its own individual set of four pistons as well as a streamlined system for connecting them to one another.

Within Double Acting Stirling Engine, the dynamic characteristics' transient behavior, which is connected to the thermal properties' behavior, is discussed. As a consequence of this, it is essential to determine the changes that have occurred over time in the thermal parameters of the working fluid. Due to the fact that the thermodynamics behavior in each cylinder cannot be treated as a distinct system, a Double Acting Stirling Engine with four cylinders may be distinguished from a Stirling engine with a single cylinder in a number of important ways. The linking mechanism brought about a change in the thermodynamic characteristics of all four cylinders, which ultimately brought about this outcome.

III.III Solar Stirling Engine

In 1987, a patent was issued for an invention by Roelf J. Meijer. [16] He proposes combining a solar dish collector with a heat engine, such as a Stirling cycle engine, to generate power. [16] A large dish in the heart of the device collects and focuses sunlight. Concentrated solar energy powers Stirling cycle engines[16], which operate by heat transfer from a hotter source to a colder sink. A generator is powered by the work output from the Stirling cycle to generate electricity [2]. Meijer's solar-powered engine requires that the solar dish collector be pointed squarely towards the sun at all times. To far, solar Stirling systems have shown to be the most efficient solar-based power producing technology [17], turning around 30% of the sun's light into electrical energy.

III.III.I Solar Stirling Engine Design

Dish for collecting sunlight The Stirling system's parabolic collector, engine, and generator are all concentrated at the dish's focal point. [18], The diagram is shown in Fig. 2

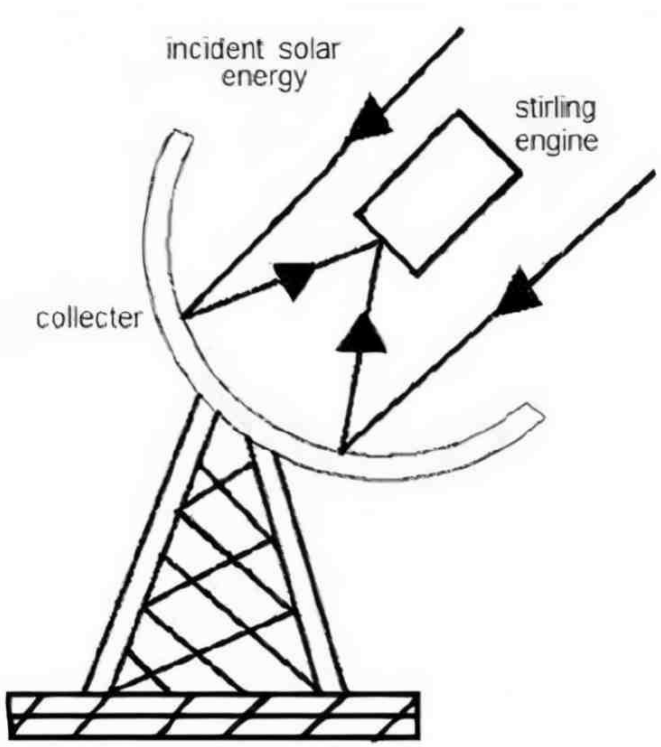


Fig. 6

Solar concentrators and thermal receivers make up the collector system. The collector system is designed to generate the thermal energy used by the Stirling engine. Thanks to the parabolic dish, the receiver aperture is focused on by the sun's beams. The receiver consists mostly of an aperture and an absorber. The absorber gathers energy from the sun, which is then transferred to the working gas in the Stirling engine via the dish's aperture. Finally, the Stirling engine converts the thermal energy into mechanical energy for use in generating electricity [19]. The model of a working dish-Stirling engine is shown in Fig. 3.



Fig. 7

III.III.II Solar Stirling Engine Performance

In this paragraph, we take a look at the latest advancements that have been achieved in the technology of solar Stirling engines and analyze the performance and efficiency of these engines. Research in modeling, design, optimization, and simulation are the primary focuses of these works. Taking into consideration the design features of the collector system, as Gholamalizadeh and Chung[19] did, led to improved results from a trial program for a solar parabolic dish Stirling system. This was done in order to generate more energy from the sun. In addition, the researchers analyzed their findings taking into consideration both exergy and energy losses. The concentration ratio was 625, and the aperture diameter of the parabolic collector on the trial dish was 0.12 meters. This dish is also sometimes referred to as the (Kerman pilot parabolic dish). A comparison between the Kerman Pilot and the Euro dish is shown in the studies. The overall efficiency of the Kerman Pilot was calculated to be 17%, whereas the efficiency of the Euro dish was judged to be 19%. [17,19]

Ahmadi and Hosseinzade [20] investigated the efficiency of a heat engine based on Stirling cycles that was fueled by solar energy. In addition, it was determined how each design option affected the temperature of the receiver as well as its overall performance. In addition to analyzing the influence of collector design characteristics, Ahmadi and Hosseinzade investigated how the total temperature transfer rate, the concentration ratio, and the heat engine parameter impacted the efficiency of solar Stirling.

Bataineh [21] modified a dish-Stirling design with the intention of increasing the engine's output power while also enhancing its efficiency. It was assumed that both the regenerator losses and the thermal bridge losses had been taken into account during the whole of the investigation. An analytical model of the solar Stirling engine's dish collector could be constructed thanks to a finite-time thermodynamic study that was performed. According to their estimates, the highest possible level of thermal efficiency could be achieved with a coefficient value of 1,300 and an absorption temperature equal to 850 degrees Kelvin (K). The results of their investigation brought them to this conclusion.

When Li Z, Tang D, Du J, and Li T [22] investigated the flux distribution on the surface of the receiver, they made use of Monte Carlo ray tracing. The study team looked at a variety of different combinations for the concentrator and the receiver to see which ones worked best. During the course of the research, both the ideal paraboloid concentrator and the faceted concentrator were analyzed and contrasted. Within the receiver cavity, fluid flow and accompanying heat transfer were modeled numerically using a CFD simulation. The radiation flux profiles served as boundary conditions for the simulation. The Monte Carlo approach and the CFD method both had their validity validated by the trials, which also proved their reliability. The inquiry also led to the creation of a novel design for a receiver assembly that maintains a consistent temperature throughout and achieves a high degree of efficiency that is equal to or more than 60.7%.

III.III.III Remarks of Solar Stirling Engine

The dish-Stirling system had efficiencies of 17% for both energy and exergy, resulting in a total efficiency of 19% for the system. The thermal efficiency of the system is most impressively shown by the value of 32%. By using a system that included a receiver and a Stirling heater, it was possible to achieve an efficiency level of 60.7%.

IV. Conclusion

The Stirling cycle engine may use several different types of fuel, including liquids, gases, and solids. The engine's design hinges on its ability to use low-temperature heat sources such solar radiation, industrial effluents heat, and heat generated from agricultural waste.

The comparison also analyzes the maximum of thermal efficiency of the dish-Stirling system, which is higher than any of the regular build, and as for the efficiency of the receiver-Stirling heater configuration, demonstrating that the Solar Stirling Engine is superior to the other two types of Stirling engines (regular and double acting) and to the three types of Stirling engines (Alpha, Beta, and Gamma). These results suggest that the Solar Stirling Engine might be a viable choice for renewable energy because of its low cost and simple

design for generating power.

New materials and enhanced heat transmission to the working fluid are crucial for the development of the Stirling engine. It is common practice to either employ a less viscous working fluid or increase the pressure to achieve the large mass fluxes necessary for efficient heat transfer. Some promising new results from research on solar-powered Stirling engines have been discovered. Though inefficient, the Stirling engine is both reliable and inexpensive. A good Stirling solar generator should be simple to operate and have consistent results.

Both the dynamic behavior of the engine mechanism and the functioning of the heat exchangers contribute to engine reliability and efficiency. It's not easy to increase a system's performance while keeping in mind the interplay of thermal, hydraulic, and mechanical design factors. Many scientists and engineers have acquired critical data for the design of engine systems since the creation of the engine. However, in order to build systems in tandem with a thermo-fluid-mechanical method, additional in-depth expertise is required. The driving mechanism and engine combination are two key factors in determining the vehicle's overall performance. The Stirling engine has the potential to become a practical alternative to conventional power plants with further improvement, such as the selection of an acceptable design, the adoption of a suitable working fluid, and the development of a superior seal.

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